Progress slides



EPFL

Investigating Retrieval Strategies For Global Health and Humanitarian Documents



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LiGHT Laboratory for intelligent Global Health & Humanitarian Response Technologies.

Week 1 Monday

Onboarding and set objectives :

- Check links Fabrice sent and make sure I have access to them (drive, presentations, git repos)
- Fresh ubuntu install on my laptop
- Do rcp tutorial
- Run mmore
- Go through mmore documentation + code
- Check with Lorin what issues to take
- First step of literature review:
    - Read papers of interesting research areas in retrieval strategies for rag at the moment
    - Write my problem statement and research question
    - Iterate

Week 1 Thursday

Done

- Access to resources
- Computer setup (fresh ubuntu with enough memory)
- Docker and rcp tutorial
- Read code and doc first sweep
- Run mmore: Rag remaining

Remaining

- Check with Lorin what issues to take
- First step of literature review:
  - Read papers of interesting research areas in retrieval strategies for rag at the moment
  - Write my problem statement and research question
  - Iterate
- Fresh mmore clone to reproduce the anthropic overloaded issue and add test

Week 1 Thursday

```
    from .decisions import step_record
  File "/lightscratch/users/aouame/mmore/.venv/lib/python3.11/site-packages/mmore/rag/judge/decisions.py", line 7, in <module>
    from .types import JudgeConfig, JudgeDecision, JudgeResult
  File "/lightscratch/users/aouame/mmore/.venv/lib/python3.11/site-packages/mmore/rag/judge/types.py", line 7, in <module>
    from ..llm import LLMConfig
  File "/lightscratch/users/aouame/mmore/.venv/lib/python3.11/site-packages/mmore/rag/llm.py", line 12, in <module>
    from langchain_anthropic import ChatAnthropic
  File "/lightscratch/users/aouame/mmore/.venv/lib/python3.11/site-packages/langchain_anthropic/__init__.py", line 4, in <module>
    from langchain_anthropic.chat_models import (
  File "/lightscratch/users/aouame/mmore/.venv/lib/python3.11/site-packages/langchain_anthropic/chat_models.py", line 1252, in <module>
    class AnthropicOverloadedError(anthropic.OverloadedError, ModelAPIError):
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: module 'anthropic' has no attribute 'OverloadedError'
```

```
● mmore ❯ python -c "import anthropic; print('anthropic:', anthropic.__version__); print('OverloadedError:', hasattr(anthropic, 'OverloadedError'))"
  anthropic: 0.73.0
  OverloadedError: False
```

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Week 2 Monday

Done

- Checked with Lorin what issues to take, i assigned myself to dependabot issues and :
  - restarting moore lose job progress (medium)
  - /retrieve w/ empty fileIds searches the whole DB unrestricted (urgent)
  - support of distributed processing for Slurm (pas urgent)
  - redo the PDF deep processor (with images) (pas urgent)
  - multimodality for final LLM (pas urgent) (voir avec Lorin)
  - Integrate nemotron (pas urgent)
- Merged pr Fix anthropic dependency issue #362
- Submitted pr Update uv lock file and enforce uv lock check #365
- Became more familiar with the codebase, rcp and docker, spent maybe too much time to understand the workflow
- Read Jeremy thesis (half)

Next

- Start literature review :
  - Read papers of interesting research areas in retrieval strategies for rag at the moment
  - Write my problem statement and research question
  - Iterate
- Start from a survey paper then branch out from there
- Have a first draft by Thursday

Week 2 Thursday

Done

- Started literature review and a first draft of problem statement + research questions
- Looked more in mmore code
- Reading papers:
  - Surveys about rag and best practices.
  - Take away:
    - Hybrid-based search then re-ranking works well for text. Tables, scans, other languages and multi-page questions are still hard
    - Choosing question from query alone is an active research area but the gain does not look certain. Deciding after a first search is also done

Collected papers:

- Chunking methods (semantic, adaptive) and their effectiveness
- Adaptive post retrieval vs pre retrieval
- Agentic rag systems
- Visual document retrieval: text vs page images

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What i am thinking for now:

- Core question: How much better is the best setup per question than the best single setup? If that gap is small, adaptivity wouldn't be worth it probably. Ask Mallory to get a set of questions with gold retrieved documents and measure that.
- Adaptive pre-retrieval: Given documents indexed with mmore and a query, can a system decide which retrieval configuration should be used based only on the query (chunking, which indexing, retrieval strategy, text or text+image)
- Adaptive post-retrieval: (~=enhance the judge) Same question but allow a first run then rerun with different configuration or choose stopping = no answer in documents
- Multimodal retrieval: When does visual retrieval or fusion of modalities beat text retrieval. Can a system predict it given a query?

Next:

- Work on one of my issues
- Continue reading and rewriting problem statement
- Which direction do you think fits mmore best? Do you recommend to avoid reading about any of the areas i collected papers from? Avoid chunking because semester project students are already doing it

Week 3 Monday

Done

- Narrowed scope after thursday feedback
- Reorganised my papers.
- Read 2 papers on adaptive post retrieval
- CRAG (Yan 2024):
    - A custom rule with thresholds splits them into 3 categories: keep and filter / web search / both.
    - It does not change the retrieval setup but only the rewrite the query + adds web insights.
    - Their evaluator beats prompted ChatGPT at judging (84% vs 58-65%). CRAG beats RAG in every setting. For PubHealth 39 vs 75.6 accuracy.
- Beyond the Query (Li 2026):
    - Label per query: answer score after the extra step minus before (ex add image)
    - Compared 2 identical routers, one sees query features only, one sees query + retrieval statistics.
    - No reliable gain according to them

What i am thinking for now:

- Crag did not try to vary the retrieval setup
- Second paper suggests that pre retrieval signal should be strong enough or close to post retrieval signals, good first a priori, i think it's worth to try both

Next:

- Dependabot PRs (10 open):
    - Docker bumps first
    - setuptools, chonkie, transformers
    - pymilvus and milvus-lite 3.0 last, they touch indexing and retrieval
    - Is passing the CI tests enough? No, launch gpu tests on rcp + add tests if i have ideas
- Continue reading and rewriting problem statement
- Reduced capacity this week, back pain, housing, i should get better after moving out by the end of the week